

Lab #7: Implementing Mux Circuits and Counters Using SimUAid

This lab covers three topics. First, you will be forming functions in SOP form using MUXes. Second, you will analyze a 2-bit Johnson counter (which will be useful in Labs 8-9). Third, you will design and build a 3-bit sequential circuit that behaves as a counter. **This lab is to be done individually.**

Part 1: Building Function in SOP form using 8-to-1 Mux

For this part, we will focus on building the function below:

$$F(A,B,C) = \prod M(1,2,4) \cdot \prod D(0,7)$$

Note that the given function is in **maxterm notation** and that don't-cares are the same in both minterm and maxterm expressions.

More information can be found in “Lab 7 Counter Example.pdf.” Also, an example circuit build has been attached for you to download and study.

Questions:

1. In your own words, define what a MUX is and how it can be used in digital logic design.
2. Design a circuit that implements the function above by using a MUX. Include a figure with a schematic of your design.
3. How would you implement a function that has 4 variables with 3-variable MUXes?
4. How many unique functions could you implement with a 5-variable MUX?

Part 2: Analyzing a 2-bit Johnson Counter

The Johnson counter—also known as a twister ribbon counter—is usually implemented as shift register with inverted feedback. The figure below illustrates a 2-bit example in SimUAid using two sequential D flip-flops.

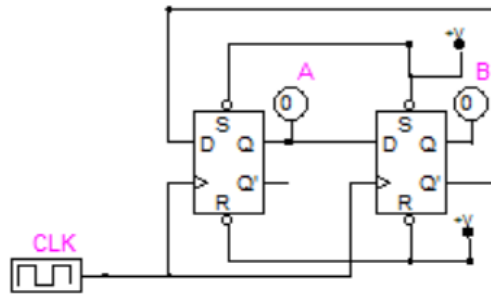


Figure 1: A Johnson counter reset to $AB=00$.

Note that $A = Q_A$ and $B = Q_B$.

Questions:

5. Find the equations for A^+ and B^+ as a function of A and B .
6. Make a transition table using your equations for A^+ and B^+ .
7. What is the period of this machine? Are there any unused states?

Part 3: Building a Sequential Counter

Now, we focus on building more complex counters using D flip-flops. You will be designing the counter based on the sequence shown in Figure 2.

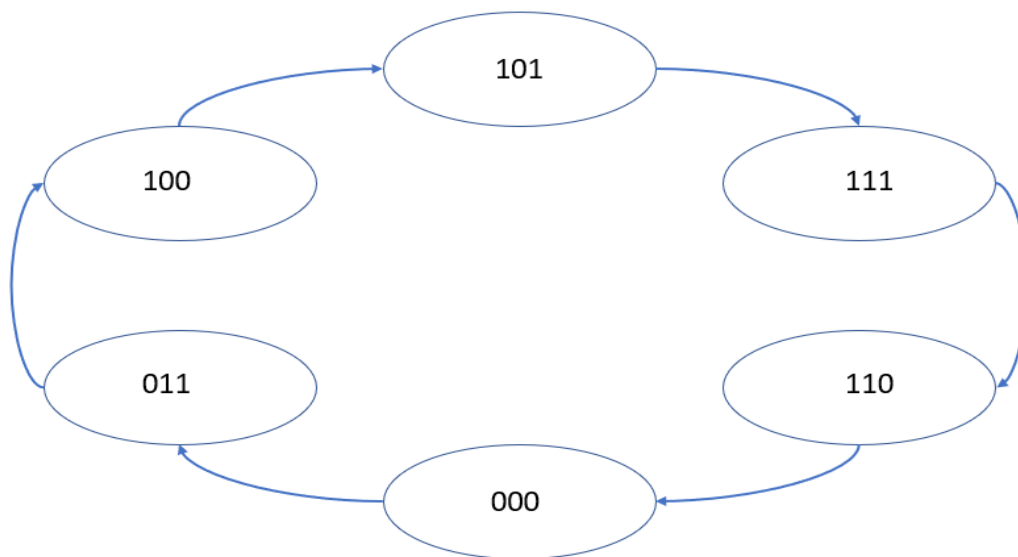


Figure 2: Sequential Counter

You will be using 3 D flip-flops. Use a new simulation file to design the counter. Attached to the lab 7 folder is a solved example and a circuit build in SimUAid. Make sure you study the timing diagram, and make sure that your circuit only uses the values given in your counter. Any output that does not follow the state graph in Figure 2 will indicate that an error has occurred. Testing your circuit design in simulation is recommended.

Questions:

8. As you can see from Figure 2, each stage is made up of 3-bit values. How many states can the machine start in?
9. Make a transition table for your sequential counter.
10. Which states are not used? Using your transition table, determine what happens if the counter starts in these states by accident.
11. If unused states accidentally create a secondary loop when they are treated as don't-cares, what can be done to fix it?
12. Provide next-state equations for your design.
13. What are the reasons for the D flip-flop having the inputs labeled as S and R? Why must we keep them high? Note: The S and R inputs should not be mistaken for the SR latch or SR flip-flop.

Part 4: Counter Circuit Hardware Implementation

Build the circuit that you designed in the previous part. You will need a clock source, so be mindful of your logistics. Buttons are very bad clock sources. The undergraduate lab has square wave generators. Alternatively, an Arduino can also produce a clean square wave from its I/O pins. If you have the components required to make a debouncing circuit, a button may also work.

Demo your lab to a DLD TA or record a video according to the guidelines below.

Questions (please research #14 before building your circuit):

14. Explain in your own words what “button bouncing” is and what it can do to sequential circuits.
15. Did you demo to a TA, or did you take a video? Describe who observed your demo or leave your link here.

Demo notes:

Make sure that you put your name somewhere visible on your breadboard. Make sure you state your name in the video.

It is required that your video is continuous, i.e. there should be no pauses, edits or interruptions to the recording. However, you may speed it up as appropriate.

The demo must show you DISASSEMBLING your circuit in your video at the end of the demo. Failure to do so will result in a loss of 50% of your grade.

The demo must be no more than a 6-minute video. Add it to your Google Drive or YouTube and put a *publicly accessible* link to that video in your report.

Part 5: Summary of Deliverables

Turn in your report in either PDF or Word format, and make sure it is named:
ECE3441_StudentID#_Lab7.

****Note: Please refrain from using notebook paper or writing on the attached lab. Make a separate Word document and have it submitted as a word or as a pdf document in BB. Points will be deducted if you do not follow instructions.**

Use the following structure for your lab report:

Course Name and Number – Semester

Lab Number and Title

Name

Overview: Goals of the Lab

Briefly (~ 1 small paragraph) outline the goal of this lab, as well as the work done.

Step-by-Step Procedures and Results

This must be in chronological order of how Lab 7 should be completed. Make sure you include all the necessary information that is mentioned in the lab and answer all questions. This should be a well-formatted report with distinct sections and question responses, not a block of contiguous text.

Conclusion

Summarize what you have learned in this lab.